, with zero resistance. Motor resistance is ; its no-load current is

. For a steady mean cargo flow (mass conveyed per unit time) of ,

find the mean motor current. Apart from cargo-belt friction and Joule heat,

assume that all other energy losses are independent of the cargo load on the belt.

III. (points) A light thin rigid rod pivots on a smooth horizontal axle through ,

freely in a vertical plane. Balls and , threaded on it, are point masses and .

Static friction has coefficient . The rod is horizontal at rest; balls and

are next to ends and , respectively. Release the system from rest. Find:

the exit position of ball , as the rod's angle to the horizontal at ball 's exit;

the exit position of ball , as the rod's angle to the horizontal at ball 's exit.